

Exploring Toronto's Air Quality and Temperature Through Data Visualization

Recently, Toronto experienced some of the worst air quality in the world. The area was covered in smoke, turning the sky a yellow-orange appearance. Several factors contributed to these conditions, but the main cause was smoke from wildfires burning across northern Ontario. At the same time, Toronto was in the middle of a heatwave, which likely worsened the situation. Hot, stagnant air can limit the movement of pollutants away from the lower atmosphere, allowing them to accumulate closer to the ground.

Although wildfires are a natural part of many ecosystems, the warmer temperatures in the region increasingly relate to climate change and may lead to more frequent and severe wildfires.

This situation prompted me to explore the links between temperature, wildfires, and air quality in Toronto. I used data visualization with the .NET version of LightningCharts to examine how air pollution levels have changed over time and if any noticeable patterns exist between pollution and temperature.

Looking at PM2.5

For this project, I am specifically interested in wildfire-related air pollution and how it has potentially varied over the years in relation to temperature. Wildfires are a large cause of releasing a variety of pollutants, including carbon monoxide, nitrogen dioxide, and ozone. However, one of the most important pollutants associated with wildfire smoke is fine particulate matter, or PM2.5. These are tiny solid particles and liquid droplets that can penetrate deep into the lungs and enter the bloodstream.

I used the Government of Canada's **National Air Pollution Surveillance (NAPS)** database for data on pollutants in the area. NAPS monitors and records pollutant concentrations at locations throughout Canada. I filtered the database to obtain PM2.5 measurements from Toronto between the beginning of 1997 and the end of 2024.

The data was provided as a .csv file, with an entry for every hour during the period:

```
Pollutant,NAPS_ID,Date_yyyy_mm_dd,Start_Time_Local_Standard,End_Time_Local_Standard
,Concentration,Units,Method_Code
PM2.5,060403,1997-01-01,0:00,0:59,8.0,µg/m3,706
...
```

This resulted in a very large dataset containing hundreds of thousands of individual measurements.

Turning the Data Into a Readable Chart

I used the .NET version of **LightningCharts** to visualize the data. First, I downloaded LightningCharts' Interactive Examples and found one that I considered a good foundation to start on.

Initially, I attempted to display the data as it was in the .csv file; hourly. However, with data covering 27 years, the chart was extremely difficult to interpret. There were simply too many points to identify meaningful long-term patterns.

To solve this problem, I created a function that groups the data by month and calculates the average PM2.5 concentration for each month. I also created a second series showing the yearly average concentration. This reduced the amount of information on the graph while still preserving the larger trends in the data making it much easier to read.

One useful feature I found of LightningCharts was its DataCursor, which I enabled with:

```
_pmChart.ViewXY.DataCursor.Visible = true;
```

I could hover over points on the chart and see the corresponding values. This made it much easier to interact with the data and understand what each point represented.

Adding Temperature Data

After creating the PM2.5 visualization, I added a second chart underneath it to examine the temperature of Toronto over the same period.

For this data, I used the Government of Canada's historical climate database. The temperature data was more complicated to work with because the database contains numerous weather stations, each with different periods and formats of recorded data.

Ultimately, I used data from two Toronto stations. The first, **TORONTO LESTER B. PEARSON INT'L A**, contained monthly temperature data from 1997 to 2013. This could easily be integrated into the chart since it was already in monthly format like my previous one.

From 2013 onward, I used data from the **TORONTO CITY** station. This dataset contained daily temperature measurements, with separate files for each year which I gathered and placed into a single file from 2013 through 2024.

Since the PM2.5 chart uses monthly averages, I wanted the temperature chart to follow the same format. This meant that for the TORONTO CITY data, I needed to create a function to group the daily measurements by month and calculate the average temperature. This would allow the two charts to share a similar axis making comparisons between temperature and air quality much easier.

More Ways to Explore the Data

Once I got the main information I wanted displayed, I added several additional features to make the charts more interactive and informative.

First, I added a third chart below the temperature chart using a `BarSeries` instead of a `PointLineSeries`. This is a bar chart that displays the average PM2.5 concentration for each month across the same 1997–2024 period.

I also added data for two additional pollutants: ozone (O_3) and carbon monoxide (CO). These pollutants are displayed alongside PM2.5 on the first chart, allowing their lines to be compared over the same period.

Additionally, I added a legend that allows each pollutant series to be turned on or off. This helps the viewer to focus on one pollutant at a time or compare several of them simultaneously.

Finally, I added some finishing touches with additional styling, colours, and labels to make the charts look nicer.

What Does the Data Show?

The data shows that the warmer months consistently have higher concentrations of pollutants. There are several possible reasons for this. Temperature and atmospheric conditions can influence how pollutants disperse and accumulate, while warmer months can also come with an increased risk of wildfire activity.

However, these graphs alone cannot establish that temperature directly causes higher pollution levels. Many factors can influence air quality, including wildfire activity, wind, precipitation, atmospheric conditions, traffic, industrial emissions, and other sources of pollution. The charts are best looked at as a way of identifying patterns and relationships that could be investigated further.

Something that surprised me is that there does not appear to be a clear long-term increase in pollutant concentrations around Toronto over the 1997–2024 period. Despite increasing concern about wildfire activity and climate change, the overall pollutant levels do not simply rise throughout the years, according to this data.

Final Thoughts

This project showed me how powerful data visualization can be when working with large environmental datasets. Starting with thousands of individual measurements, I was able to create a handful of interactive charts that make long-term patterns much easier to see and explore. `LightningCharts .NET` provided the tools I needed to turn raw data into something more accessible and interactive.